ARU Validator — Guide

The ARU Validator lets you listen to short clips that the HawkEars classifier flagged for one species, and say whether the species is really there. Each species has its own validation database (a `.db` file). The app reads the clips listed in it, plays them, and writes your answer back into the same file the moment you press a key.

The app only validates. It never creates, copies, renames or deletes databases or clips.

1. Install and update

You need R (version 4.0 or later) and an internet connection. Open R (RGui or RStudio), paste this line into the console and press Enter:

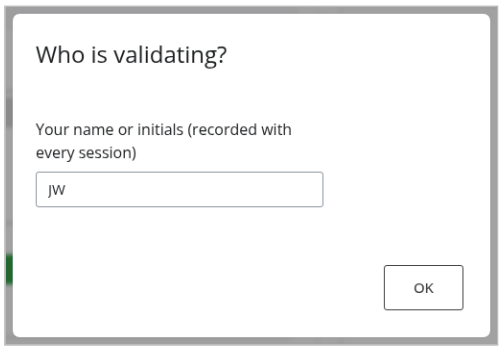
```
source("https://raw.githubusercontent.com/tabanid/aru-validator/main/install_validator.R")
```

- The first time, a window asks where to put the app; a folder **ARU-validator** is made there. Later runs update that same folder without asking.
- Missing R packages are installed (a few minutes, once).
- On Windows an **ARU Validator** shortcut is put on the Desktop. Double-click it to start the app; a black window opens beside the browser. Closing it stops the app.
- To update, paste the same line again. Updates replace only the app's own files, never your databases, clips or settings.

After an update, press **Ctrl + Shift + R** once in the browser so it loads the new version.

2. Start and say who you are

When the app starts it asks for your name or initials. They are stored with every session in each database you open (with the app version), so it is clear later who validated what. The app remembers the name for next time.



3. Open a database

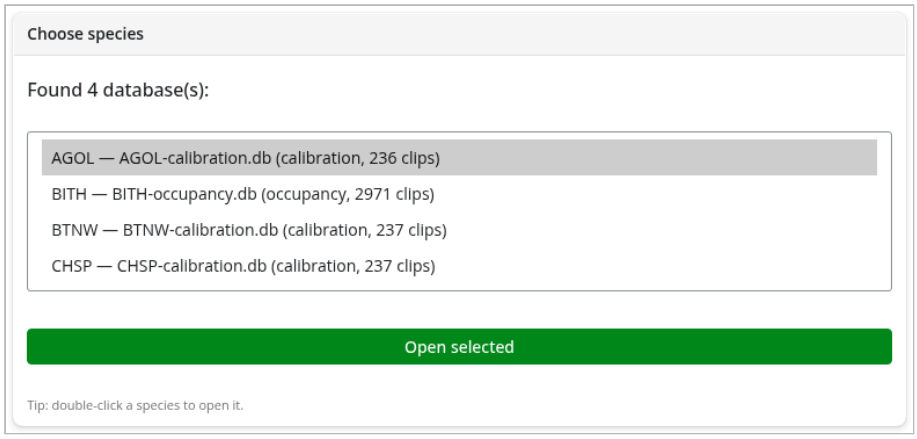
In **Open validation database** you can type either:

- **a folder that holds species folders** (for example `C:\Contracts\2026\tabanid\Tobeatic-rare`), then click **Open / Scan** (or press Enter).

The app lists every database it finds; double-click one (or pick it and click **Open selected**); or

- **the full path of one `.db` file**, then click **Open / Scan**.

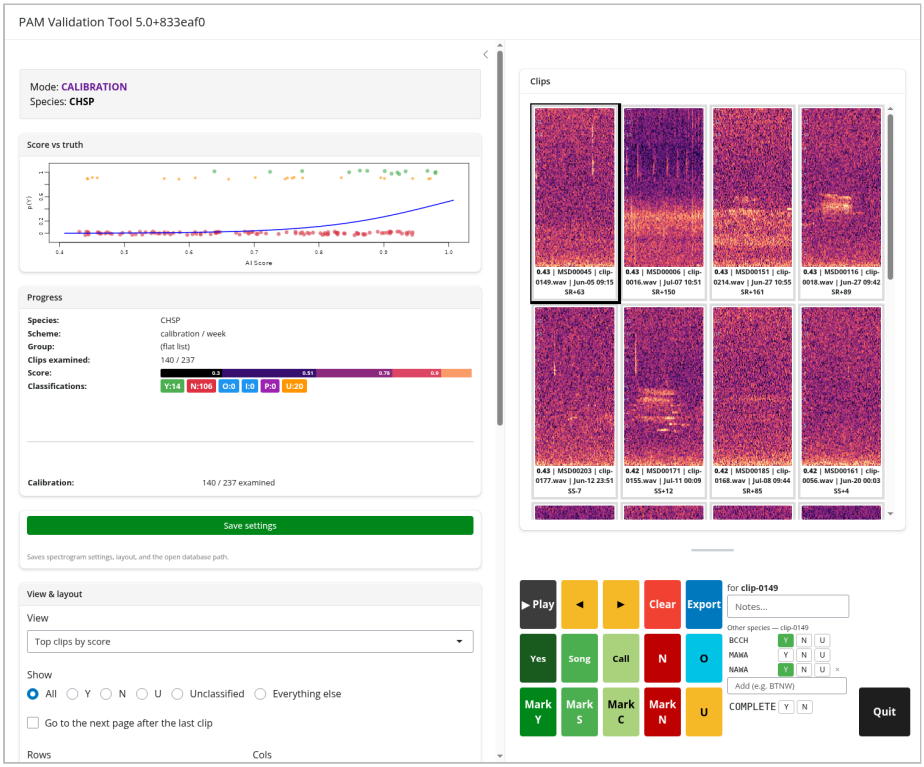
Under the box is a list of folders: click into one, or use **↑ Up** and **Home**, and the box follows. The app remembers the last folder: next time it starts there with the species already listed. The panel always says which database is open.



The app opens only databases made for this version. An older one is listed with **needs conversion** and refused with a message; nothing is written to it. Send it to Phil, who converts a copy.

The app finds each clip's sound file by looking, in order: in the same folder as the database; in a `clips/<species code>/` folder next to the database's folder; and in the folder recorded inside the database when it was made. You do not need to set anything.

4. The screen



- **Left panel:** what is open (mode, species), the score-vs-truth plot, progress, settings, and the database picker.
- **Clips:** one spectrogram per clip, highest score first. Under each: the classifier score, the recorder (unit), the clip file, and the date and time (with minutes from sunrise or sunset, e.g. *SR+9*). A clip's border takes the colour of its label; the selected clip has a thick black outline.
- **Buttons:** three rows under the clips. Top: Play, back/forward, Clear, Export. Middle: Yes, Song, Call, N, O. Bottom: Mark Y, Mark S, Mark C, Mark N, U. Notes and the Other species list sit to the right of all three rows; Quit is at the far right of the bottom row. Everything has a key (see the cheat sheet), so the buttons are optional.

5. Label clips

Click a clip (or use the arrow keys) to select it, press **Space** to play it, then press a code:

Key	Label	Meaning
Y	Y	the target species (**S** = song, **C** = call, if you can tell)
N	N	not the target
U	U	unsure

The label is saved at once and the next clip is selected. After the last clip on a page the app stays on that clip; press ► for the next page (or tick **View & layout → Go to the next page after the last clip**). **O** takes you to the Other species list (section 6); it labels nothing. The **I** and **P** buttons are switched off: insects and peepers go in the list as `Insects` and `FROG-PEEP`. **X** (or **Clear**) removes a clip's label, its notes and its other species, if you change your mind. **Alt + a code key** gives every unlabelled clip on the screen that code, for a screen that is all noise, say.

Clicking a clip that is already selected sets where playback starts (a thin line); Space then plays from there.

6. Other species and notes

Under Notes is the **Other species** list for the selected clip: every other species HawkEars scored on that clip, one row each, in code order. Each row has **Y** (heard), **N** (not heard) and **U** (unsure). Leave a row blank if you did not listen for it. The list always belongs to the selected clip and changes when you move.

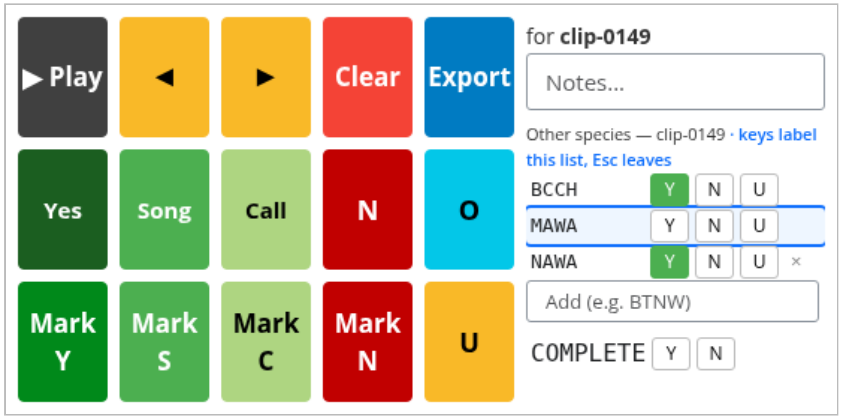
From the keyboard:

- **O** puts the highlight on the first row. The heading then says *keys label this list*.
- **↑ ↓** move between rows; **y**, **n**, **u** label the highlighted row. Pressing the same key again clears it.
- **a** goes to the **Add** box: type species HawkEars missed (for example `BTNW NAWA`, or `squirrel`, `frog-peep`; spaces or commas, any case) and press **Enter**. They are added as heard (Y). An added species has a **×** to remove it.

- Under the Add box, **COYO**, **RESQ**, **PEEP** and **INSE** (coyote, red squirrel, peeper, insects) add that code as heard with one click. They do not label the clip.
- **Other payoff**, the line at the bottom, says whether the Other list is still worth the time. Each Y you add this session counts as *useful* if it saves a clip in another species' database in the same folder (that week is not done there yet), or if it is a detection nobody would otherwise have (HawkEars missed the species that week, or every clip there was a N). It shows how many of your last 20 were useful. When none of the last 20 were, the line turns yellow: what you hear is already known, and the Other list is adding little.
- **Space** plays the clip; **Esc** or **Enter** leave the list.

While you are in the list, **y n u** label the highlighted species, never the clip, and the other keys (S, C, X, arrows between clips) do nothing. So one clip goes: **O**, **↓**, **y**, **Esc**, then **Y** (or N, U) for the clip itself.

With the mouse, click **Y N U** on a row (click a lit one again to clear it). Under the Add box, press **Y** on the **COMPLETE** row when every species you heard on the clip is labelled or added, **N** if not (click a lit one again to clear it). Every click is saved at once and never moves to the next clip, and the list can be used before or after the clip's own label. A code the app does not recognise is shown in amber. It is still saved.



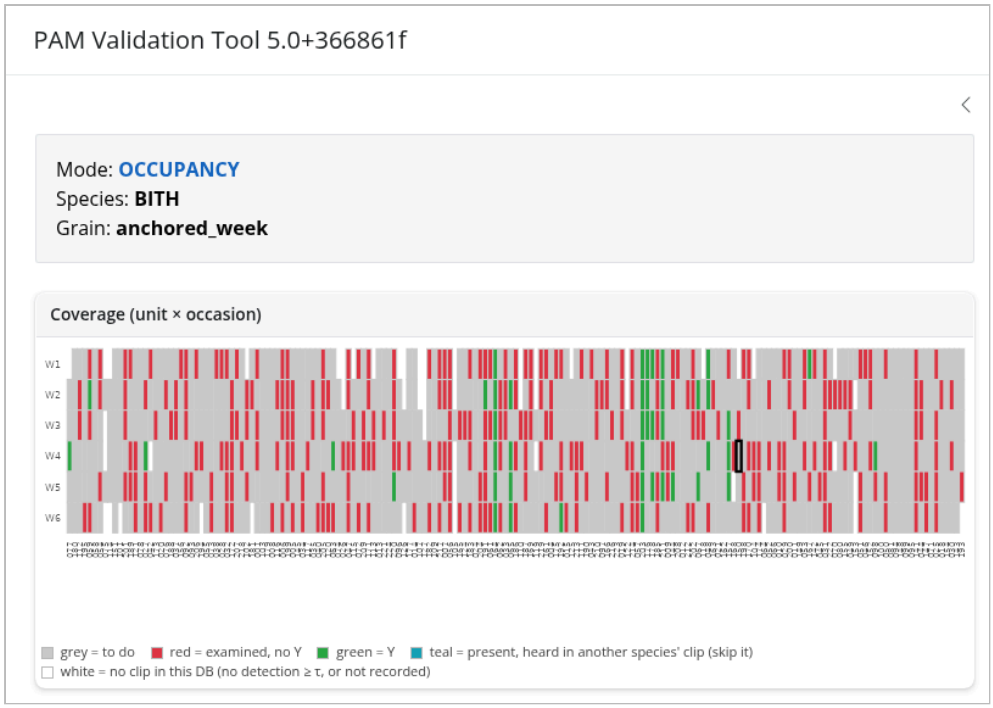
Notes is for anything else (a faulty microphone, "listen to the longer recording"). The line above it names the clip it belongs to (*for clip-0012*). **Enter** saves and stays on the clip; **Esc** undoes what you typed. While the cursor is in Notes or the Add box, letters are text, not shortcuts.

7. Two kinds of database

The mode is shown at the top of the left panel.

Calibration (Sample 1). Every clip is to be labelled. These answers are used to set each species' score threshold. The app opens on **Top clips by score**: the unlabelled clips, highest score first. **Reload** brings up the next batch.

Occupancy (Sample 2). Clips are grouped by recorder (unit) and week. One Y is enough to show the species was there that week, so you only need to work until the first Y in each group. A group with every clip examined and no Y is an absence.



- The **Group** box at the top shows the current unit × week and whether it is done. **Next unresolved** jumps to the next group that still needs work.
- The **coverage plot** has one square per unit (columns, west to east, labelled with the last digits of the recorder number) and week (rows, the first week at the bottom): grey = to do, red = N, no Y, yellow = U, no Y, green = Y found, teal = heard in another species' clip, white = no clip in this database for that unit and week. The square you are on has a black outline. **Click a square** to open that group; hover to see its name.
- A **teal** square is already done: someone marked this species **Y** in the Other species list of another species' clip from that unit and week (in Sample 1, the BITH census, or Sample 2). The app skips it, and hovering names the clip it came from. Click it to listen anyway. **Progress** counts these apart from your own Y's.

Tip — validate common, vocal species first in Sample 2. They turn up most often as "other" in other clips, so doing them early donates the most skipped cells to the species you do later. Order changes only your time, never the final result.

8. Views

Choose under **View & layout → View**:

View	Shows
Top clips by score	unlabelled clips, highest score first
Best clip per cell	the best unlabelled clip from each unit × week without a Y (occupancy)
Occupancy groups	one unit × week at a time (occupancy)
All clips (review)	every clip, labelled or not
All clips for one unit	every clip from one recorder, across all weeks, highest score first

Show (under View) narrows any view to one kind of label: All, Y, N, U, Unclassified, or Everything else (older I, P and O labels). With **All clips (review)** it is the way to go back over your answers, for example every U. The two work-queue views (Top clips, Best clip per cell) hold unlabelled clips only, so choosing Show there switches to **All clips (review)**.

The view only changes what is on screen; labels are the same whichever view you use. A database where every clip is already labelled opens on **All clips (review)**.

9. Settings

- **Spectrogram settings** (bottom of the left panel): frequency range, contrast, window, colours. Click **Update spectrograms** to apply.
- **Rows / Cols / Clip height** set how many clips are on a screen.
- **Show p(correct)**: the pipeline's estimate of how likely each clip is correct. It is off by default and is for information only; label what you hear.
- **Save settings** keeps these for next time.

If the screen is too small: drag the grey bar above the buttons up to shrink the clip area, click **‹** at the top of the left panel to hide it, or zoom the browser out with **Ctrl** and **−**. The keys work without the buttons.

10. What is saved, and where

Everything is saved straight into the database you opened. For each clip the app writes only: the label, whether it has been examined, song or call, the time, and the notes; and the Other species list (one row per species with its Y, N or U, and the *COMPLETE* answer). It also adds one line per session recording who validated and with which version of the app. **Export** writes a CSV of the labels next to the database, with the other species in one column (for example `REVI:Y;SWTH:N`).

11. If something goes wrong

Problem	Try
"cannot open URL" when installing	check the internet connection and paste the line again
a package will not install	close other R sessions, paste the line again
a key does nothing	click on a clip (the cursor may be in a text box or the list), or press Esc
a database says "needs conversion"	it was made for the old version; send it to Phil
no sound	click Play once with the mouse (browsers block sound until you click)
the old version still shows	press Ctrl + Shift + R in the browser
spectrograms say "audio not found"	the clips are not beside the database or in the folder it expects; tell Phil where they are